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Department of Computer Science and Engineering  
Program: Bachelor of Science in Computer Science and Engineering

## **Project Final Report**

Course No: CSE-1200

Course Title: Software Development – I

Project Title: Echoes of Arcana

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Submitted to

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# Introduction

## 1.1 Project Overview

Echoes of Arcana is a 2D fantasy adventure game developed in C/C++ using the iGraphics library. The player takes the role of a young student mage who must pass three trials to protect the magical academy: a maze-based challenge, a timed item-collection challenge, and a final boss battle. The game combines maze navigation, item collection, power-ups, and multi-phase combat, and it features a level-selection screen where each new level is unlocked by completing the previous one.

## 1.2 Objective

- Build unique gameplay mechanics across multiple levels.
- Implement maze navigation with collision detection, items and roaming enemies.
- Create timed item collection with dynamic difficulty scaling.
- Design a multi-phase boss fight with unique attack patterns and Rage Mode.
- Unify the core systems: movement, HP, Mana and combat.
- Distribute tasks among the 4 members and track progress through checkpoints.

# Game Development

## 2.1 Tools & Technologies Used

- **Programming Language:** C, C++
- **IDE:** Microsoft Visual Studio
- **Graphics Library:** iGraphics (built on OpenGL and GLUT)
- **Image Loading:** stb\_image.h and a custom bitmap\_loader.h for sprites and backgrounds
- **Audio:** background music and sound effects triggered on in-game events
- **Platform:** Windows PC

## 2.2 Core Features

- **Title Screen and Level Select:** The game starts with a title screen and a level selection screen. Later levels remain locked until the previous trial is completed.
- **Level 1:** Labyrinth of Whispers: The player moves through a maze, collects coins and books, avoids enemies, and uses wall collision. The level also tracks score and lives. The player starts with 3 lives, and reaching the exit gate shows a victory screen with the total score (N for Level 2, R to replay).

- **Level 2: Potions and Spells:** The player must collect 10 wands and 10 books before time runs out. Items are generated randomly with different probabilities and three difficulty phases. Shield, Magnet, time bonus and time penalty items can also appear, along with the Cursed Book, which slows the player and lowers the book count.
- **Level 3: Final Trial Arena:** The level has three phases: spiders, dark minions, and the Arcane Guardian. The boss uses magic attacks, has a separate HP system, and enters Rage Mode during the fight.
- **Player Systems:** The player can move with WASD or arrow keys and uses HP, Mana, spells, and a dash or dodge move.
- **HUD and Game States:** The HUD shows HP, Mana, item counters, time and power-up information. Each level also has victory and failure states with retry, next-level and menu controls (R, N and M). In Level 3, R restarts from the phase where the player fell.
- **Characters and Enemies:** The player has four-direction sprites (down, up, left, right) and a separate spell effect. Level 3 has three enemy types: the Spider (Phase 1, chases the player), the Dark Minion (Phase 2, melee and ranged attacks) and the Arcane Guardian (Phase 3 boss with magic attacks and Rage Mode).
- **Items and Power-ups:** Level 1 has coins (raise the score), books (collectible) and the exit gate. Level 2 has falling wands, book stacks and golden spellbooks, with Shield (temporary protection), Magnet (pulls items in), time bonus, time penalty and the Cursed Book.

## Implementation Details

### 3.1 Code Structure

The project is split across several files rather than one large main file, which made it possible for four people to work on different parts at the same time without constantly stepping on each other's code:

- **iMain.cpp** – Main entry point. It sets up the iGraphics window, controls the main draw and update loop, and switches between the title screen, level select and active level. It also contains the Level 1 code: the maze, item placement, enemy movement, wall collision, and lives and score tracking.
- **level2.h / level2.cpp** – Level 2 (Potions and Spells) code: item spawning, difficulty phases, item effects, timer, keyboard input and HUD.
- **pujita\_level3.h / pujita\_level3.cpp** – Level 3 arena, enemy and boss logic, phase changes, attack patterns, Rage Mode and HP handling.
- **bitmap\_loader.h / stb\_image.h** – Used to load and draw sprite, background and UI image assets.

- **iGraphics.h** – Provides the graphics, timer, keyboard and mouse input functions used by the project.
- **level2\_assets, level3\_assets** – Folders containing the image assets for Level 2 and Level 3.

### 3.2 Challenges Faced & Solutions

| Challenge                       | Solution                                                                                                                                                                                                               |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Balancing Level 2 item spawning | Early tests sometimes produced too many items or too few items. The team adjusted spawn probability and falling speed for each difficulty phase and limited the number of items on screen during Magical Chaos Mode    |
| Balancing the Arcane Guardian   | The boss fight could become too easy or too difficult depending on the phase. HP thresholds, attack cooldowns and Rage Mode were adjusted through repeated testing.                                                    |
| Working on shared files         | Changes could be overwritten when four members edited shared files without a continuous shared Git workflow. The team assigned file ownership for each checkpoint and compared builds manually before merging changes. |
| Item collision boxes            | Some Level 2 item images did not match their collision areas after the artwork was changed. The collision box of each item was adjusted separately.                                                                    |

# Screenshots

The following screenshots are taken from the finished build of Echoes of Arcana.



Figure 1: Title Screen



Figure 2: Level Select Screen

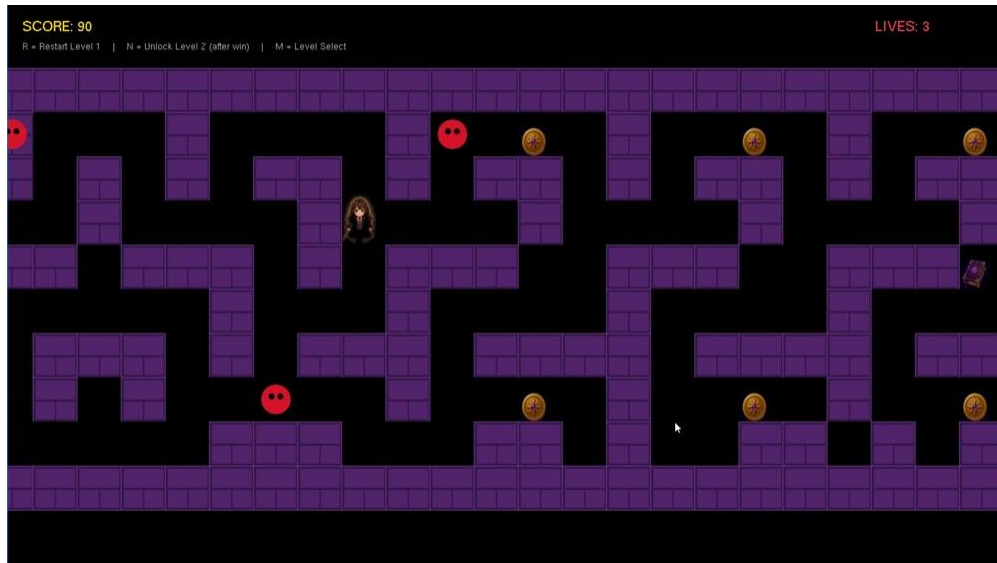


Figure 3: Level 1 – Labyrinth of Whispers (collecting coins, avoiding enemies)

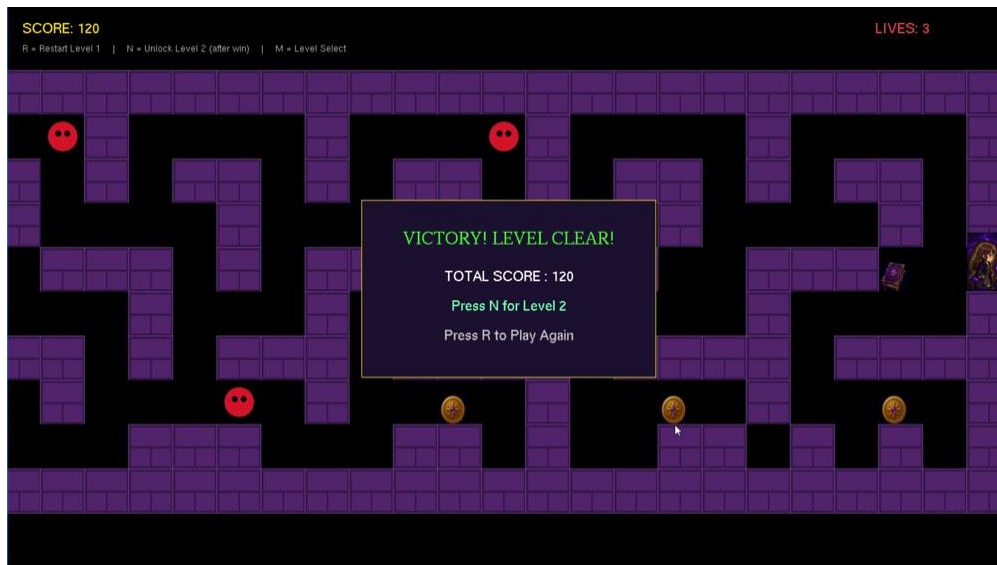


Figure 4: Level 1 – Victory Screen



Figure 5: Level 2 – Story Cutscene

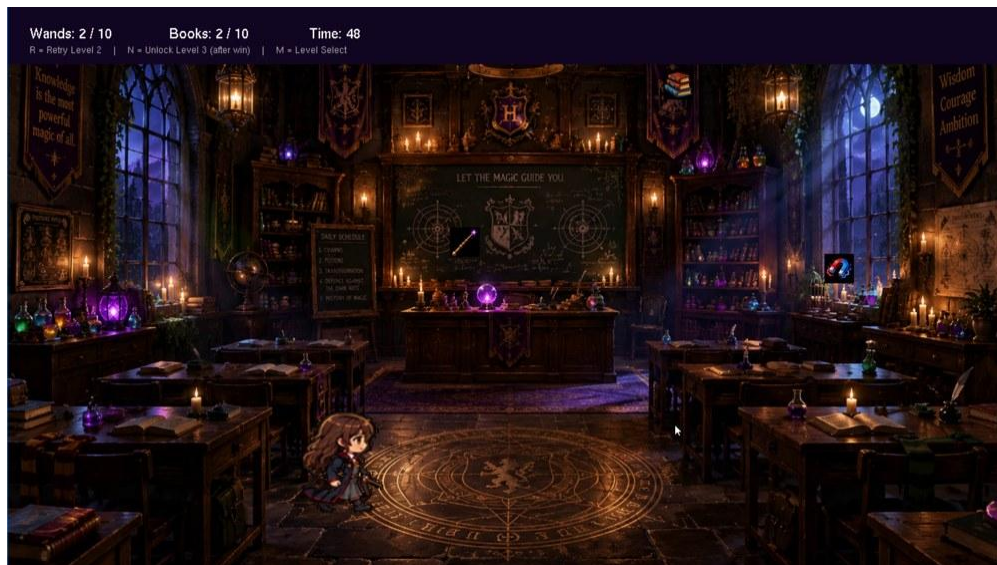


Figure 6: Level 2 – Potions and Spells (collecting wands and books)

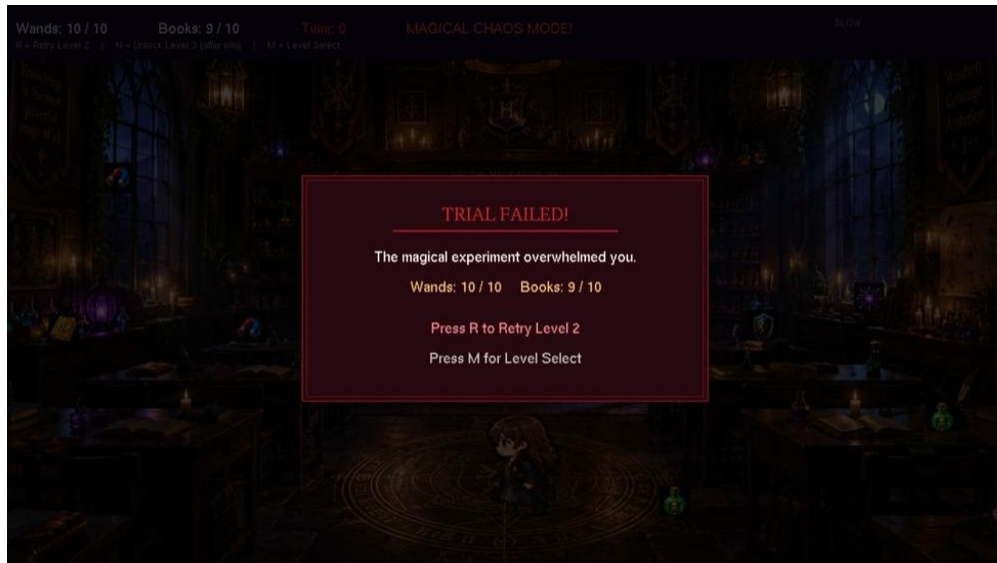


Figure 7: Level 2 – Trial Failed Screen (Magical Chaos Mode)

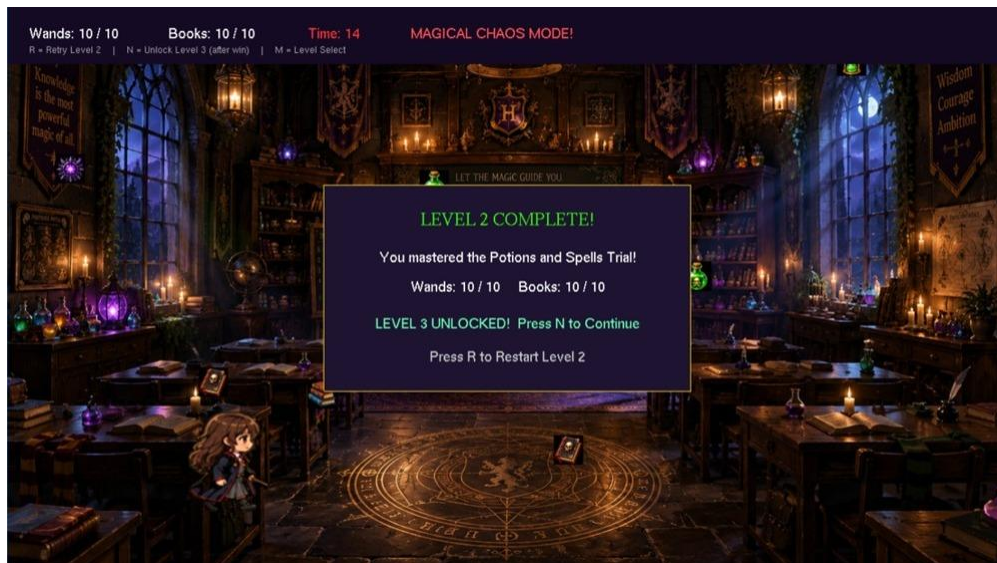


Figure 8: Level 2 – Completion Screen and Level 3 Unlocked



Figure 9: Level 3 – Phase 1: Spiders



Figure 10: Level 3 – Phase 2: Dark Minions



Figure 11: Level 3 – Phase 3: Arcane Guardian Boss Fight

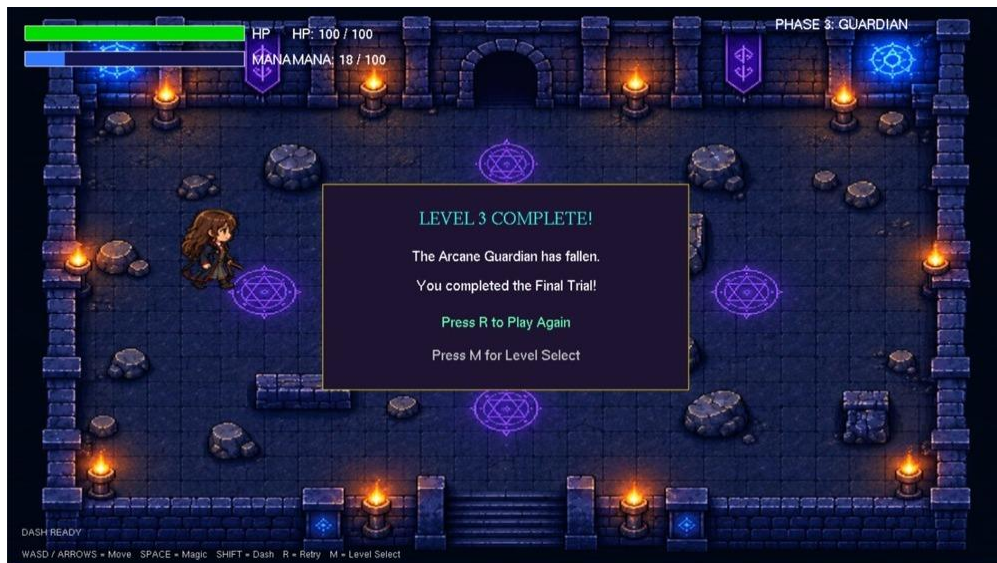
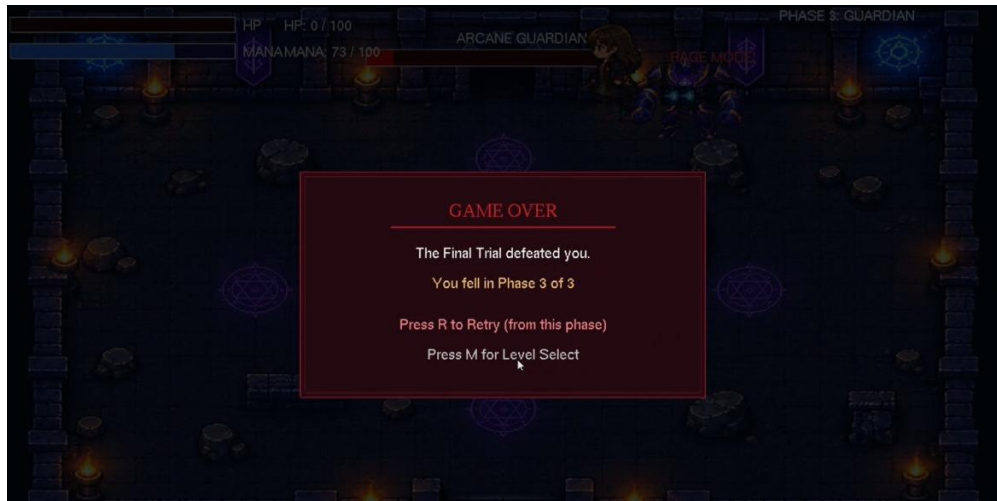


Figure 12: Level 3 – Completion Screen (The Arcane Guardian has fallen)



**Figure 13: Level 3 – Game Over Screen**

## 4. GitHub Collaboration

### 4.1 Repository Management

Rather than working off one continuously updated git history, the team tracked progress through three checkpoints spaced across the semester, with each checkpoint report documenting who built what and how much of that stage's work belonged to each member. Now that the game is finished, the completed project is being uploaded to a GitHub repository for submission and as a version reference going forward.

GitHub Repository: <https://github.com/pujitapodder44-ui/Echoes-of-Arcana>

### 4.2 Contribution Workflow

Across the three checkpoints, the team worked roughly like this:

1. At the start of each checkpoint, the scope for that stage was agreed on (e.g. Level 1 in checkpoint 1, Level 2 in checkpoint 2, the boss fight and enemy system in checkpoint 3).
2. Work was split by feature area rather than by file, so each member owned a specific system for that stage.
3. Members built their part independently and merged it in manually once it was working.
4. At the end of each checkpoint, contributions were written up and reported as a percentage split, based on what was actually built.

This kept the workload roughly balanced across checkpoints and meant everyone knew what they were responsible for before each stage started, even without a shared branch and pullrequest workflow running the whole time.

### 4.3 Individual Contributions

Summarised across all three checkpoints:

- **Pujita Podder (00725105101154)** – project setup and integration. Set up the base project and main menu in the first checkpoint, merged everyone's work and handled early bug fixes, and put together the background music. Built the Level 2 item-spawning system, the difficulty phases, falling-item speed, and weighted random generation behind what drops. In the final checkpoint, built the Arcane Guardian boss fight from scratch: the three phase structure, attack patterns, Rage Mode, the magic attacks and HP system, and the win logic once the boss is defeated.
- **Debi Podder (00725105101157)** – narrative and presentation. Wrote the backstory in the first checkpoint. Handled Level 2's core game logic -collision detection, item-effect handling for every drop type, the timer countdown, and win/lose state management. In the final checkpoint, worked on game assets and presentation: character and enemy art, arena visuals, UI elements, and integrating the story images and sound assets.

- **Shahara Jebin Raisha (00725105101168)** – player facing mechanics across all three levels. Drew the Level 1 maze and placed the coin, book and exit-gate assets, and handled player movement and wall collision. Built the Level 2 keyboard input layer - movement, boundary handling, and the restart/next-level shortcuts. In the final level, owned player control and core gameplay: movement, HP, Mana, spellcasting, dash/dodge and the damage system.
- **Nabonita Saha (00725105101169)** – enemy systems and interface work. Handled player placement, the enemy system, score/lives tracking and sound effects in the first checkpoint. Built the Level 2 visuals and HUD - sprite rendering, the wand/book/time counters, chaos-mode and power-up indicators, and the win/lose and level-unlock screens. In the final checkpoint, built the Level 3 enemy system: spiders and dark minions, their movement and chase behaviour, HP, and melee and ranged attacks.

## 4.4 Contribution Evidence

Since progress was tracked checkpoint by checkpoint rather than through a long-running git history, the contribution percentages reported at each checkpoint (summarised above) are the main record of who built what and how much. The finished project has also been pushed to the GitHub repository linked in section 4.1 for submission.

# Challenges & Limitations

## 5.1 Challenges

The biggest technical challenge was not any single feature but keeping three very different levels working together inside one program. The player, the HUD, and the win/lose logic are shared across the maze, the falling-item level, and the boss fight, while each level has its own rules, timers, and screens. Because of this, a bug in one place often showed up as a strange problem somewhere else. For example, a state that was not reset properly when we switched levels made the score, timer, or health carry over wrongly into the next level. Adding the R, N, and M shortcuts for retry, next level, and level select also meant every level had to reset itself cleanly. Chasing these cross-level bugs took longer than writing the original features.

iGraphics itself was also a source of friction. It is a small library built for course use, so much of its behaviour had to be found by trial and error. It worked reliably only with 24-bit BMP files, our images appeared misaligned at first, and getting transparency needed a black-background workaround with iShowBMP and iShowBMP2. When we combined our files, iGraphics.h was included in more than one place and caused a linker error (LNK2005), which we fixed with extern declarations. Some Level 2 item images also stopped matching their collision areas after the artwork was changed, so we had to adjust the collision box of each item separately.

Balancing the difficulty was the last big challenge. In Level 2, early tests produced either too many or too few falling items, so we tuned the spawn probability and falling speed for each

difficulty phase and limited the number of items on screen during Magical Chaos Mode. In Level 3, the Arcane Guardian could feel too easy or too hard depending on the phase, so we adjusted HP thresholds, attack cooldowns, and Rage Mode through repeated playtesting. On top of this, four of us worked on shared files without a continuous Git workflow, which meant changes could be overwritten. We solved this by assigning file ownership for each checkpoint and comparing builds manually before merging.

## 5.2 Limitations

- **Combat stays within spells and basic attacks:** there is no weapon-upgrade system or item shop beyond the power-ups that appear mid-run in Level 2.

Apart from this, the planned features are complete: three levels, rising difficulty, the boss fight with Rage Mode and the shared player systems. The game can be played from start to end.

# Conclusion & Future Improvements

## 6.1 Summary

Echoes of Arcana ended up as three trials that genuinely play differently from each other - a maze to navigate, a timed collection challenge that gets harder as it goes, and a three-phase boss fight with its own Rage Mode built on iGraphics and split across four people over three checkpoints. Keeping each level's systems separate while still sharing the same player, HUD and win/lose logic underneath was most of the actual work, more than any single feature on its own. File handling is also used to keep the score consistent across levels.

## 6.2 Future Enhancements

- **More Levels & Bosses** – extend the story past the current three trials with new mechanics.
- **Difficulty Options** – let the player choose how aggressively Level 2's chaos mode and the boss's Rage Mode scale.
- **Advanced Animations** – more frames for spell casting and the Arcane Guardian's attacks.